

## Lu Xiao

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## Education

- **Nanchang University** **Nanchang, China**  
*B.Eng. in Electronic Information Engineering* *Sept. 2022 – June 2026 (Expected)*
    - **Honors College:** Jiluan Academy (Honors Program for Innovative Talent, Top 1% of University).
    - **GPA:** 3.46/4.00 (Weighted Avg: 85.21/100) | **Rank:** 2/7 (Honors Class).
    - **Selected Coursework:** Advanced Mathematics (94/100), Python Programming (90/100), Circuit Analysis (92/100), Digital Systems & Logic Design (I) (90/100), Digital Systems & Logic Design (II) (97/100), Theoretical Mechanics(94/100).
  - **Tsinghua University & X-Institute** **Shenzhen, China**  
*Long-term elite student training program* *Jan. 2025 – Present*
    - Selected for the long-term elite student training program modeled after Tsinghua's Tsien Excellence in Engineering Program (TEEP).
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## Honors & Awards

- **Second Prize**, Nanchang University Innovation Contest 2024
  - **Third Prize**, Lanqiao Cup National Software and IT Talent Competition (National Qualifiers) 2025
  - **Excellent Student Scholarship**, Nanchang University 2022-2023
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## Publications

- **A Multimodal Depression Diagnosis Model Inspired by Traditional Chinese Medicine's Observation and Inquiry**  
Haijun Che, **Lu Xiao** (Co-First Author), Enhao Gu, Xinqi Fan, Shiguang Ni.  
*2025 19th International Conference on Complex Medical Engineering (CME)*, IEEE, pp. 411–415, 2025.  
**DOI:** 10.1109/CME67420.2025.11239373
  - **Decoding the Reactivity Enhancement of  $\text{Ln}_2\text{Ce}_2\text{O}_7$  Compounds (Ln = Yb, Y, Tb, and Gd) for Soot Combustion**  
Shijing Zhang, Junwei Xu, Zekai Xu, Yuting Li, Ping Wang, **Lu Xiao**, Jiating Shen, Xiuzhong Fang, Xianglan Xu, Xiang Wang.  
*Inorganic Chemistry*, American Chemical Society, vol. 63, no. 15, pp. 6798–6812, 2024.  
**DOI:** 10.1021/acs.inorgchem.4c00142
  - **Effect of Sunlight on Long-term H59 Brass Corrosion Monitoring Model Using Deep Learning**  
**Lu Xiao**, Yao Wang, Guanhao Chen, Zhiqin Wu, Haofeng Zhang, Chao Luo, Honggun Song, Zhi hu.  
*Manuscript in Preparation (Targeting Corrosion Science)*.
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## Research Experience

- **Tsinghua University (AI & ROBOT Lab)** **Shenzhen, China**  
*Research Intern* (Advisor: Prof. Xueqian Wang) *Mar. 2025 – Sept. 2025*  
**Project: Safe Planning for Robotic Arms Based on Goal-conditioned Reinforcement Learning**
  - Developed a robotic grasping policy using the **DDPG** algorithm, integrated with the **Hindsight Experience Replay (HER)** framework for sparse reward environments.
  - Designed simulation experiments involving dynamic obstacles to test safe planning capabilities.
- **Tsinghua University (Positive Psychology & Serious Games Lab)** **Shenzhen, China**  
*Research Intern* (Advisor: Assoc. Prof. Shiguang Ni) *Mar. 2025 – Sept. 2025*  
**Project: Multimodal Depression Diagnosis Model Inspired by TCM**
  - Proposed a multimodal diagnosis model fusing PHQ-9 scale data and facial features to reduce misdiagnosis.
  - Extracted facial features using **DeepFace** and constructed an ensemble model (based on CoE framework) integrating expert classifiers.
  - Led data processing, feature extraction, and co-wrote the conference paper.
- **Nanchang University (Advanced Manufacturing Laboratory)** **Nanchang, China**  
*Research Intern* (Advisor: Prof. Zhi Hu) *Oct. 2023 – Feb. 2025*  
**Project: Long-term H59 Brass Corrosion Monitoring Model**
  - Developed a monitoring method fusing Electrochemical Noise (EPN) and GLCM image features.
  - Optimized signal analysis using **Wavelet Transform** and established a **NARX neural network** for corrosion rate prediction.
- **Nanchang University (Key Lab of Environment & Energy Catalysis)** **Nanchang, China**  
*Research Intern* (Advisor: Prof. Xiang Wang) *Sept. 2022 – Sept. 2023*  
**Project: Reactivity Enhancement of  $\text{Ln}_2\text{Ce}_2\text{O}_7$  Catalysts for Soot Combustion**
  - Synthesized a series of  $\text{Ln}_2\text{Ce}_2\text{O}_7$  ( $\text{Ln} = \text{Yb}, \text{Y}, \text{Tb}, \text{Gd}$ ) nanocatalysts via glycine combustion method.
  - Characterized physicochemical properties using **XRD**, **Raman**, and **XPS** to analyze the impact of variable valence A-sites on crystal structure and surface oxygen defects.
  - Investigated the catalytic mechanism and confirmed that variable valence A-sites (especially in  $\text{Tb}_2\text{Ce}_2\text{O}_7$ ) significantly enhance soot oxidation activity and  $\text{NO}_x$ -assisted efficiency.

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## Skills

- **Programming:** Python, C++, Verilog, MATLAB, LaTeX
- **AI & Robotics:** PyTorch/TensorFlow, ROS, Reinforcement Learning (DDPG), Computer Vision
- **Hardware & EDA:** Multisim, Proteus, Keil, Altium Designer